

Query Run Report

Research Paper Synthesis Graph · arm: typed_graph

1 Run Configuration

Field	Value
Question	<i>“Why is community detection hard and how can quantum computing help?”</i>
Retrieval arm	<code>typed_graph</code>
Invocation	<code>scripts/ask.py --system typed_graph</code>
Embedding model	<code>sentence-transformers/allenai-specter</code>
Graph index	25,410 node names embedded
Compute device	mps (Apple Silicon)
Synthesis model	<code>gpt-5.4-mini</code>
Papers grounded	4
Wall clock	≈ 64 s

Hub requests were unauthenticated (no `HF_TOKEN` set); this affects download rate limits only, not retrieval or results. Model weights resolved from cache — 199 shards, sub-second load.

2 Timeline

From	To	Elapsed	Stage
19:30:09	19:30:12	≈ 3 s	Device selection, encoder load (cached)
19:30:12	19:31:09	≈ 57 s	Typed-graph node-name embedding (25,410 names)
19:31:09	19:31:13	≈ 4 s	Synthesis call
Total		≈ 64 s	

Node-name embedding accounts for roughly **89% of wall clock**. It is a whole-graph cost that does not vary with the query, so it belongs in a persisted index rather than in the request path — the single largest available latency win for this arm.

3 Synthesized Answer

Why community detection is hard

In multipartite networks the problem is “not trivial” and “usually requires ad hoc methodologies”.¹ Modularity maximization has “been largely studied” with many competing methods proposed — itself evidence that this is a difficult optimization problem rather than a straightforward task.² Performance is characteristically evaluated on benchmark graphs as mixing increases, indicating sensitivity to graph structure.³

¹9be428c9

²19e9eaa0

³9be428c9

How quantum computing can help

- Community detection can be recast as an optimization problem that a **quantum annealer** can solve.⁴
- One paper presents a **modularity-based formulation for multiple communities** and runs it on the **D-Wave** annealer.⁵
- The reported result is a “highly optimized community structure”, with the annealer achieving the best value based on community structure among the methods compared.⁶
- More generally, quantum computers are described as a means to solve problems intractable in reasonable time classically, with quantum algorithms potentially outperforming classical ones on specific problems.⁷

Stated limitation

The system flagged its own evidence gap: the retrieved excerpts *do not* supply a single unified explanation of *why* community detection is hard in general. They establish three narrower points only — nontrivial structure, ad hoc handling in multipartite settings, and modularity maximization as a challenging optimization task.⁸

4 Grounding

Four distinct papers were cited. Full Semantic Scholar identifiers:

Short	Full identifier	Role in answer
15eded04	15eded04386a8982ccd5627bd1efe70bbf624c02	Quantum advantage, general
19e9eaa0	19e9eaa00a36ed3334ea0c18b337a18d18f95c3e	Modularity QUBO, D-Wave
9be428c9	9be428c9383d47b86570b1b9fc20faf006346c5d	Multipartite hardness
f9db7ae0	f9db7ae0a333ef8a21317d1a3126d75da9d43ff4	Quantum advantage, general

Every claim carries a paper-level citation, and the two halves of the question draw on different sources — hardness from 9be428c9, the quantum result from 19e9eaa0 — rather than resolving onto one dominant paper.

5 Cost

Model	Calls	In	Out	Cached	Cost
gpt-5.4-mini	1	1,194	301	0	\$0.00
Total	1	1,194	301	0	\$0.00

⁴19e9eaa0

⁵19e9eaa0

⁶19e9eaa0

⁷f9db7ae0, 15eded04

⁸9be428c9, 19e9eaa0

6 Cross-Arm Context

Not from this run — carried over from the `fulltext` vector arm on the same question, for comparison.

Measure	<code>fulltext</code>	<code>typed_graph</code>
Input tokens	11,316	1,194
Output tokens	421	301
Papers grounded	1	4
Wall clock	≈ 9 s	≈ 64 s
Cost	\$0.01	\$0.00

The typed graph reached **four papers on one-tenth the input tokens**, and spread its citations across the two halves of the question — where the vector arm retrieved 20 chunks from 13 papers and grounded on one. That is the behaviour the typed graph is built to produce.

Two caveats keep this from being a result. First, one question is an anecdote, not evidence; it needs the full 34-question set with per-query-type breakdown before it means anything. Second, the typed-graph answer explicitly hedged that its excerpts gave no unified account of the hardness — so the thinner evidence budget may be buying precision at the cost of coverage. Both arms should be scored against the gold set on citation recall before either is preferred.